Problem 10: Privacy-Preserving Federated Learning System for Collaborative AI Across Multiple Institutions

Domain: Federated Learning | Distributed ML | Privacy Engineering | Model Optimization
Difficulty: Hard

Problem

Hospitals, universities and banks hold valuable data but cannot pool records due to privacy law, contracts and governance.

Objective

Build a multi-client federated learning simulator and compare collaborative training with local-only (and centralized) training, studying heterogeneity, unreliable clients and communication cost.

Users

ML engineers in multi-institution collaborations, privacy officers.

Core Requirements

- Multiple clients with separate datasets.
- Local-only baseline per client.
- Federated training (FedAvg or similar).
- Aggregate updates without centralizing raw data.
- Non-IID client data.
- Track global performance per round.
- Compare federated vs local-only vs centralized.
- Different dataset sizes and participation rates.
- Measure communication overhead and convergence.
- Dashboard of training and per-client performance.

Advanced Challenges

- Non-IID drift (FedProx, SCAFFOLD).
- Client dropout.
- Weighted aggregation.
- Differential privacy via calibrated noise.
- Optional secure aggregation.
- Malicious/malformed updates (robust aggregation: median, Krum).
- Privacy-communication-accuracy trade-offs.

Suggested Datasets

FEMNIST/LEAF, CIFAR-10 with Dirichlet partitioning, a tabular healthcare set (e.g., eICU or a public clinical dataset), Shakespeare (LEAF) for text.

Suggested Tech

Python, PyTorch, Flower, scikit-learn, Opacus, Streamlit, FastAPI.

Reference Architecture

Partitioner (IID / Dirichlet / quantity skew) → N client trainers → server (strategy plug-ins: FedAvg, FedProx, robust aggregators) → DP module → metrics logger → dashboard.

Evaluation

Accuracy/F1 (global and per client), rounds to target accuracy, bytes communicated, client fairness (variance, worst-client accuracy), privacy-utility curve (epsilon vs accuracy), attack success under poisoning.

Important Note

Keeping data local is not automatic privacy. State the threat model (honest-but-curious server, malicious clients) and test any privacy claim, e.g., with a gradient-inversion or membership-

inference attack.
Pitfalls
Only IID experiments; reporting epsilon without stating the accounting method.
Suggested Timeline (48h)
0-8h partitions + baselines; 8-20h FedAvg + metrics; 20-32h non-IID/dropout/robustness; 32-40h DP + attack test; 40-48h dashboard.

Standard Judging Rubric (applies to every problem)

Problem understanding	10%
Technical depth	20%
Data engineering (quality, leakage prevention, reproducibility)	10%
Model performance (metrics, baselines, honest evaluation)	15%
System architecture	15%
Working prototype	10%
Explainability and responsible AI	10%
Innovation and scalability	5%
Presentation and demo	5%

Mandatory Submission Checklist

- □ GitHub repository with setup instructions
- □ Working prototype or recorded demo
- □ Documented system architecture
- □ Dataset sources and preprocessing methodology
- □ Evaluation results with baseline comparisons
- □ Limitations and failure cases
- □ One technically challenging feature demonstrated
- □ Short presentation: problem, approach, results, future work